

Lecture 23

Model comparison part 1

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(Edited from the lecture notes by Brian J. Reich, NC State)



Model selection

- ▶ So far we have many potential models in our toolbox.
- ▶ For a given dataset, how to determine whether a simple model is sufficient or if we need a very complex one?
- ▶ There is probably no “right” model in practice.
- ▶ A **statistical model** is just a mathematical representation of the system that includes errors and biases in the observational process.
- ▶ All models are basically some simplifications of the reality and the reality is never that simple.
- ▶ We want a model that is as simple as possible yet seems to explain the patterns in the data reasonably well.

Outline of Chapter 5

- ▶ For model comparisons, there are a finite number of candidate models and we want to select one using
 - ▶ Bayes factors
 - ▶ Stochastic search variable selection
 - ▶ Cross validation
 - ▶ Deviance information criteria (DIC)
 - ▶ Watanabe-Akaike information criteria (WAIC)
- ▶ In cases where multiple models fit well, we will discuss
 - ▶ Bayesian model averaging (BMA)
- ▶ After selecting a model, we want to test whether it fits the data well
 - ▶ Posterior predictive checks

Bayes factors (BF)

- ▶ In some sense, BFs are the gold standard.
- ▶ Suppose, we are comparing two models, $\mathcal{M}_1$ and $\mathcal{M}_2$.
- ▶ For example, $Y \sim \text{Binomial}(n, \theta)$ and the two models are

$$\mathcal{M}_1 : \theta = 0.5 \text{ and } \mathcal{M}_2 : \theta \neq 0.5.$$

- ▶ Another example would be $Y_1, Y_2, \dots, Y_n$ being a time series and

$$\mathcal{M}_1 : \text{Cor}(Y_{t+1}, Y_t) = 0 \text{ and } \mathcal{M}_2 : \text{Cor}(Y_{t+1}, Y_t) > 0.$$

- ▶ Another example in the multiple linear regression setting would be

$$\mathcal{M}_1 : \text{E}(Y) = \beta_0 + \beta_1 X \text{ and } \mathcal{M}_2 : \text{E}(Y) = \beta_0 + \beta_1 X + \beta_2 X^2.$$

Bayes factors (BF)

- ▶ BF is basically the same as the frequentist hypothesis testing.
- ▶ We proceed by computing the posterior probability of the models $\mathcal{M}_1$ and $\mathcal{M}_2$.
- ▶ This requires prior probabilities $p(\mathcal{M}_1)$ and $p(\mathcal{M}_2)$.
- ▶ Note that this is not a prior on a parameter, rather it is a prior on the set of models!
- ▶ This approach permits conclusions like “Given the data we have observed, the quadratic model is 5 times more likely than a linear model”.

Bayes factors (BF)

- ▶ The BF for model $\mathcal{M}_2$ compared to model $\mathcal{M}_1$ is

$$BF_{2,1} = \frac{\text{Posterior odds}}{\text{Prior odds}} = \frac{p(\mathcal{M}_2|\mathbf{Y})/p(\mathcal{M}_1|\mathbf{Y})}{p(\mathcal{M}_2)/p(\mathcal{M}_1)} = \frac{p(\mathbf{Y}|\mathcal{M}_2)}{p(\mathbf{Y}|\mathcal{M}_1)}.$$

- ▶ Rule of thumb: $BF_{2,1} > 10$ is strong evidence for $\mathcal{M}_2$.
- ▶ Rule of thumb: $BF_{2,1} > 100$ is decisive evidence for $\mathcal{M}_2$.
- ▶ In linear regression, *BIC* approximates the log-BF comparing a model to the null model.

Example

- ▶ $Y \sim \text{Binomial}(N, \theta)$ with

$$\mathcal{M}_1 : \theta = 0.5 \text{ and } \mathcal{M}_2 : \theta \neq 0.5.$$

- ▶ $p(Y|\mathcal{M}_1)$ is just the binomial density with $\theta = 0.5$.
- ▶ $\mathcal{M}_2$ involves an unknown parameter θ .
- ▶ This requires a prior, say $\theta \sim \text{Beta}(a, b)$, and integration

$$p(Y|\mathcal{M}_2) = \int p(Y, \theta) d\theta = \int p(Y|\theta) p(\theta) d\theta.$$

Calculation of $p(Y|\mathcal{M}_2)$

- Under $\mathcal{M}_2$, we have

Likelihood: $Y|\theta \sim \text{Binomial}(N, \theta)$,

Prior: $\theta \sim \text{Beta}(a, b)$.

- Thus,

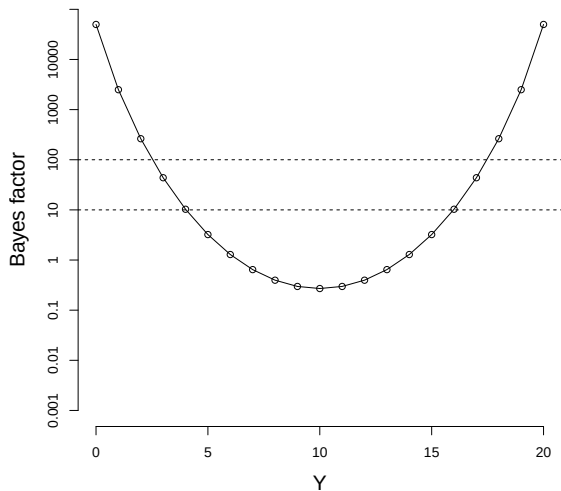
$$\begin{aligned} p(Y|\mathcal{M}_2) &= \int p(Y, \theta) d\theta = \int p(Y|\theta) p(\theta) d\theta \\ &= \int \binom{N}{Y} \theta^Y (1 - \theta)^{N-Y} \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \theta^{a-1} (1 - \theta)^{b-1} d\theta \\ &= \binom{N}{Y} \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \int \theta^{Y+a-1} (1 - \theta)^{N-Y+b-1} d\theta \\ &= \binom{N}{Y} \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \frac{\Gamma(Y+a)\Gamma(N-Y+b)}{\Gamma(N+a+b)}. \end{aligned}$$

Calculation of $BF_{2,1}$

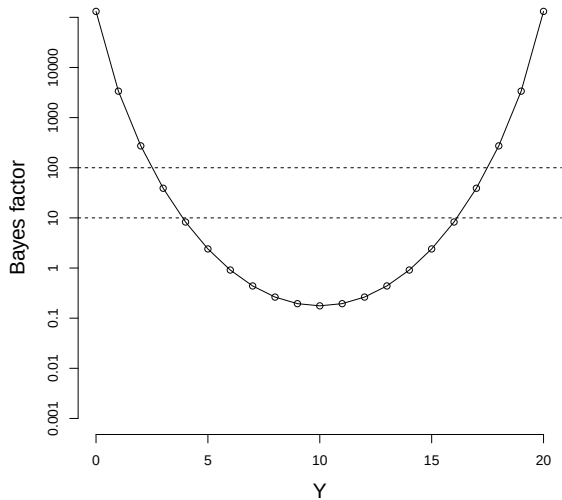


$$\begin{aligned} BF_{2,1} &= \frac{p(Y|\mathcal{M}_2)}{p(Y|\mathcal{M}_1)} \\ &= \frac{\binom{N}{Y} \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \frac{\Gamma(Y+a)\Gamma(N-Y+b)}{\Gamma(N+a+b)}}{\binom{N}{Y} 0.5^Y 0.5^{N-Y}} \\ &= \frac{\frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \frac{\Gamma(Y+a)\Gamma(N-Y+b)}{\Gamma(N+a+b)}}{0.5^Y 0.5^{N-Y}}. \end{aligned}$$

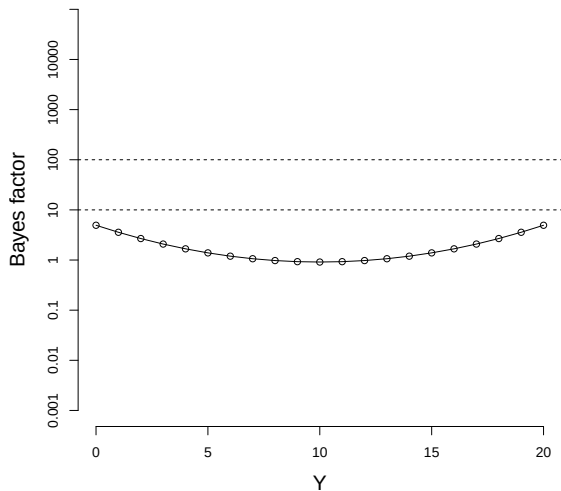
BF by Y with $N = 20$ and prior $a = 1$ and $b = 1$



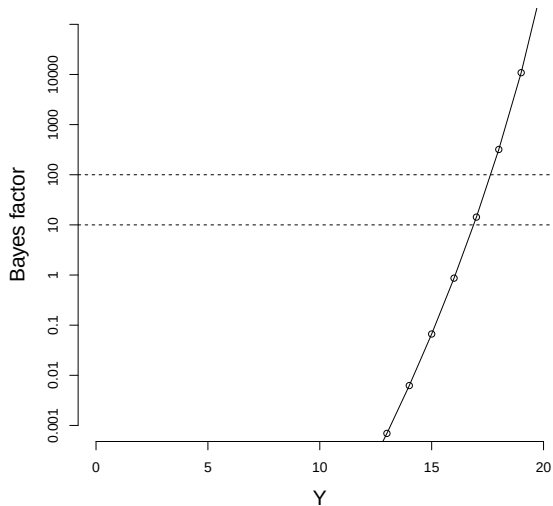
BF by Y with $N = 20$ and prior $a = 0.5$ and $b = 0.5$



BF by Y with $N = 20$ and prior $a = 50$ and $b = 50$



BF by Y with $N = 20$ and prior $a = 50$ and $b = 1$



Problems with Bayes factors

- ▶ Often hard to compute the required integrals which is only feasible for simple models.
- ▶ Requires proper priors.
- ▶ Can be very sensitive to priors (Lindley's paradox).
- ▶ In most cases, computing posterior credible intervals from the full model and then testing by comparing these to the null is preferred.

Computing Bayes factors using MCMC

- ▶ If models can be written as nested, then MCMC can be used to approximate model probabilities.
- ▶ For example, say

$$\mathcal{M}_1 : E(Y) = \beta_0 + \beta_1 X \text{ and } \mathcal{M}_2 : E(Y) = \beta_0 + \beta_1 X + \beta_2 X^2.$$

- ▶ Both model can be written as

$$E(Y) = \beta_0 + \beta_1 X + \gamma \beta_2 X^2,$$

where $\gamma \in \{0, 1\}$ indicates the model.

- ▶ The prior on models becomes $\gamma \sim \text{Bernoulli}(0.5)$.
- ▶ Then $\text{Prob}(\gamma = 1 | \mathbf{Y}) = \text{Prob}(\mathcal{M}_2 | \mathbf{Y})$ can be approximated using MCMC.

Stochastic search variable selection (SSVS)

- ▶ This is the Bayesian analog of forward/backward/stepwise variable selection.
- ▶ We place a prior on all 2^p models using p variable inclusion indicators γ_j .
- ▶ MCMC returns the approximate posterior probability of each model.
- ▶ With large p all models will have low probability and so this requires long MCMC runs.
- ▶ Similar to Bayes factor, SSVS can be sensitive to priors.

Model averaging

- ▶ Let us go back to the linear regression example

$$\mathcal{M}_1 : E(Y) = \beta_0 + \beta_1 X \text{ and } \mathcal{M}_2 : E(Y) = \beta_0 + \beta_1 X + \beta_2 X^2.$$

- ▶ Say we have fit both models and found that both are about equally likely, but that $\mathcal{M}_1$ is slightly preferred.
- ▶ For prediction, $\hat{Y}$, we could simply take the prediction that comes from fitting $\mathcal{M}_1$.
- ▶ But the prediction from $\mathcal{M}_2$ is likely different and nearly as accurate.
- ▶ Also, taking the prediction from $\mathcal{M}_1$ suppresses our uncertainty about the form of the model.

Model averaging

- ▶ Let $\hat{Y}_k$ be the prediction from model $\mathcal{M}_k$ for $k = 1, 2$.
- ▶ The model averaged predictor is

$$\hat{Y} = w\hat{Y}_1 + (1 - w)\hat{Y}_2.$$

- ▶ It can be shown that the optimal weight w is the posterior probability of $\mathcal{M}_1$.
- ▶ Averaging adds stability.
- ▶ In linear regression with p predictors the prediction is a weighted average of 2^p possible models.
- ▶ This is implemented in the R package `BMA`.

Thank you!